UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Taylor's and Maclaurin's Expansion

Complete study pack for class 007: why series are used, Taylor's and Maclaurin's series in one variable, the standard series, and Taylor's and Maclaurin's expansion for a function of two variables

Compiled from the PESU class 007 slide deck and AV-summary transcript, the Unit 1 class-work sheet and past ESA papers. Generated 14 October 2026.

· How to use this pack

Topics 1–2 are the one-variable background (a self-learning component in the syllabus); topics 3–4 are the examinable two-variable expansions. Class 007 states the series but works no problems: the lecture defers them to class 008 (“Problems on Taylor's and Maclaurin's series”), which is a separate session and is not included. The worked examples here come from the class-work sheet and past ESA papers, so you can practise every exam pattern from this pack. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

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5. Appendix A · Symbol glossary
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1 Why Taylor's and Maclaurin's series?

A complicated function is hard to evaluate, integrate or take limits of; a polynomial is easy. Taylor's series writes a function as a polynomial (of infinitely many terms) around a point, and a few terms are usually enough. This is how values such as $e^{\tan \sqrt[3]{3}}$, integrals such as $\int_0^1 \sin(x^2)\,dx$ and limits such as $\lim \frac{\sin x - x}{x^3}$ are handled, and it has many applications in physics and engineering.

Graphs and diagrams

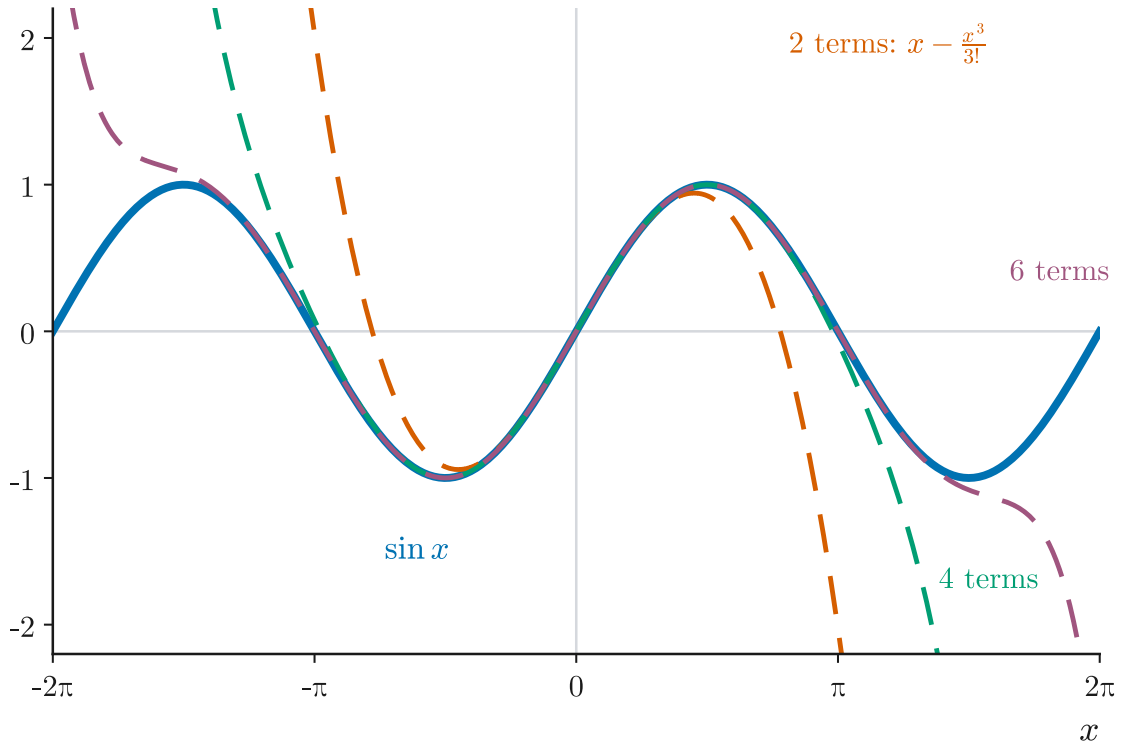


Figure G1.1. $\sin x$ and its Maclaurin polynomials. ESA ★

S007-8 · T007 00:14:45 · T007 00:15:58

How to read it. Blue is $\sin x$; the dashed curves keep 2, 4 and 6 terms of $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$. Each extra term extends the range where the polynomial follows the function: near 0 all of them agree; far from 0 the short polynomials shoot off. The slide shows 4, 6 and 10 terms; with 10 terms the curves overlap on this range.

Important points and why

1. Once a function is a polynomial, you can differentiate, integrate, find limits and study how it changes. ESA ★ T007 00:06:50

2. Applications (one variable): polynomials for complicated functions, definite integrals with no closed form, asymptotic behaviour, growth of functions, differential equations, the Lorentz factor, $E = mc^2$ at low speed, $\sin\theta \approx \theta$ for a pendulum, power-flow analysis, generating functions. ESA ★ S007·6 · S007·7 · T007 00:11:31 · T007 00:13:44
3. Applications (two variables): optimization (a linear or quadratic model), accurate distance measurement with the sine law, variance of complicated configurations, visualizing complicated functions. ESA ★ S007·12 · T007 00:21:22
4. A few terms are often enough: four terms of the sine series already follow $\sin x$ closely near 0, and ten terms overlap it (Figure G1.1). ESA ★ S007·8 · T007 00:15:24 · T007 00:16:10
5. Taylor's theorem is the basis of L'Hôpital's rule and the mean value theorems of Rolle, Lagrange and Cauchy; the series is a consequence of the theorem. ESA ★ TRANSCRIPT ONLY T007 00:03:33

Why does a calculator need series at all?
It can only add and multiply. A polynomial needs nothing else, so a function such as $e^{\tan x}$ is evaluated through polynomial approximations.
T007 00:01:02 · T007 00:01:35

Exam trap. Slide 007·3's limit $\lim \frac{\sin x - x}{x^4}$ does not exist; with x^3 the answer is $-\frac{1}{6}$ (W1.2).
ADDED FOR CLARITY S007·3

Worked examples

W1.1 · A definite integral with no closed form — ESA ★

GIVEN	Evaluate $\int_0^1 \sin(x^2)\,dx$.
STEP 1	$\sin(x^2) = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \dots$ (replace x by x^2 in the sine series).
STEP 2	Integrate term by term: $\frac{x^3}{3} - \frac{x^7}{7\cdot 3!} + \frac{x^{11}}{11\cdot 5!} - \dots$
STEP 3	At the limits: $\frac{1}{3} - \frac{1}{42} + \frac{1}{1320} - \dots$
ANSWER	≈ 0.3103
CHECK	sympy: $\int_0^1 \sin(x^2)\,dx = 0.310268$; three terms give 0.310281.

T007 00:11:51 · T007 00:12:20

W1.2 · A limit by series (slide 007·3, corrected) ESA ★

GIVEN Find $\lim_{x\to 0}\dfrac{\sin x - x}{x^3}$.

STEP 1 $\sin x - x = -\frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

STEP 2 Divide by x^3: $-\frac{1}{6} + \frac{x^2}{120} - \dots \to -\frac{1}{6}$.

ANSWER $\boxed{-\dfrac{1}{6}}$

CHECK sympy agrees. With x^4 (as printed on the slide) the quotient behaves like $-\frac{1}{6x}$ and has no limit.

ADDED FOR CLARITY S007·3

SHORT-NOTE RECAP

Series turn complicated functions into polynomials, which can be evaluated, integrated and used in limits: $\int_0^1 \sin(x^2)dx \approx \frac{1}{3} - \frac{1}{42} + \frac{1}{1320} = 0.3103$; $\lim \frac{\sin x - x}{x^3} = -\frac{1}{6}$. Applications: integrals, asymptotics, differential equations, Lorentz factor, $\sin\theta \approx \theta$, optimization, visualization.

2 Taylor's and Maclaurin's series in one variable

If f has continuous derivatives, $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$ about $x = a$; about the origin this is Maclaurin's series. The first two terms are the tangent line. The standard series for $(1+x)^n$, e^x, $\log(1+x)$, $\sin x$, $\cos x$, $\sinh x$, $\cosh x$, $\tan^{-1} x$, $\sin^{-1} x$ and $\cos^{-1} x$ are on slide 007·5. This one-variable part is a self-learning component of the syllabus.

Key terms and definitions

Taylor's series (one variable). If a function $f(x)$ has continuous derivatives of order $(n+1)$, then this function can be expressed as a polynomial at a point $x = a$ in the form $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \dots$ S007·4 · T007 00:04:19

Maclaurin's series (one variable). If the function is expanded about the origin we get Maclaurin's series: $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \dots$; it is a particular case of Taylor's series. S007·4 · T007 00:06:17

Formulas

E2.1 · Taylor's series, one variable ESA ★

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \dots$$

gives	A polynomial form of f near $x = a$.
applies	f has derivatives of every order at a and the series converges to f.
note	Slide 007·4 writes $f^n(a)$ for the n-th derivative $f^{(n)}(a)$.

S007·4 · T007 00:04:56

E2.2 · Maclaurin's series, one variable ESA ★

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \dots$$

gives	Taylor's series about the origin.
applies	f and all its derivatives finite at 0 (not, e.g., $1/x$ or $\log x$).
note	"About a point $x = a$: Taylor; about the origin: Maclaurin."

S007·4 · T007 00:11:17

E2.3 · Exponential and logarithm

ESA ★★

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad (x \in \mathbb{R})$$
$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (|x| < 1)$$

GIVES Two standard series.

APPLIES As stated.

NOTE e^{-x}: replace x by $-x$.

S007·5 · T007 00:07:19 · T007 00:08:39

E2.4 · Binomial and geometric series

ESA ★

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$
$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \qquad (|x| < 1)$$

GIVES Series for powers.

APPLIES $|x| < 1$ (any n; for a positive integer n the series stops).

NOTE The geometric series is said in the lecture, not on the slide.

S007·5 · T007 00:07:37

E2.5 · Trigonometric and hyperbolic

ESA ★★

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$
$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots, \qquad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

GIVES Odd and even standard series.

APPLIES All real x.

NOTE Slide 007·5 prints $\cos x = 1 - \frac{x^2}{2} + \dots$ ($2! = 2$).

S007·5

E2.6 · Inverse trigonometric ESA ★

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$\sin^{-1} x = x + \frac{1}{2}\frac{x^3}{3} + \frac{1\cdot 3}{2\cdot 4}\frac{x^5}{5} + \dots, \qquad \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

GIVES Series for the inverse functions.

APPLIES $-1 \le x \le 1$.

NOTE Slide 007·5 prints $\tan^{-1} x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$; the denominators are **3** and **5**, not **3!** and **5!**.

S007·5

Graphs and diagrams

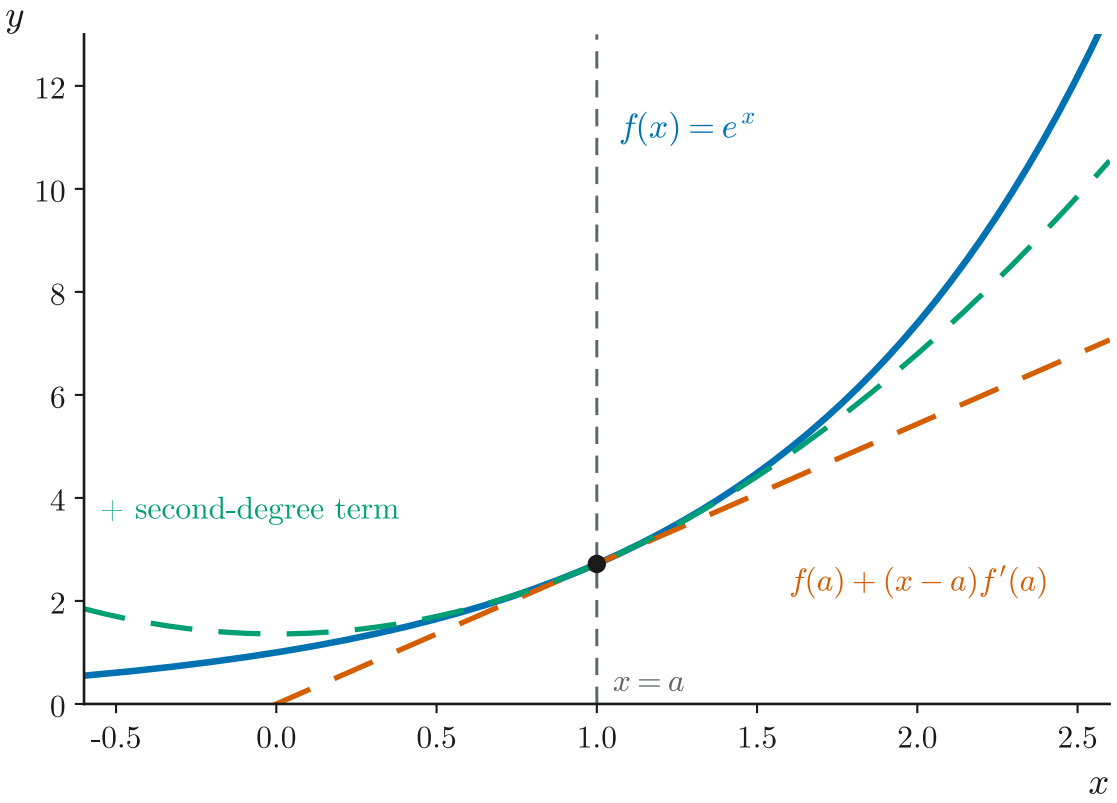


Figure G2.1. The first terms of Taylor's series: tangent line and parabola. ESA ★

ADDED FOR CLARITY T007 00:05:26 · T007 00:05:43

How to read it. $f(x) = e^x$ about $a = 1$. Two terms, $f(a) + (x-a)f'(a)$, give the tangent line (vermilion). Adding $\frac{(x-a)^2}{2!}f''(a)$ gives a parabola (green) that bends with the curve and stays close over a wider range.

Important points and why

1. Keep only the first two terms of Taylor's series and you get $f(x) \approx f(a) + (x-a)f'(a)$: the tangent line, i.e. the derivative as a slope (Figure G2.1). ESA ★ T007 00:05:26 · T007 00:05:43
2. To get a standard series, differentiate repeatedly and put $x = 0$: every derivative of e^x is e^x, which is 1 at the origin. ESA ★ T007 00:09:10 · T007 00:09:44
3. Substitute into a known series rather than differentiating: $\sin(x^2) = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \ldots$. ESA ★★ T007 00:11:51
4. No Maclaurin series exists if f or a derivative is infinite at 0: $1/x$, $\log x$, $\sqrt{x}$. ESA ★ T007 00:10:18

When is Maclaurin's series valid?
f must have derivatives of every order at 0 (so none is infinite there), and the series must converge to f. e^x, $\sin x$, $\cos x$ converge for every x; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.

ADDED FOR CLARITY T007 00:10:01 · S007·5

Exam trap. The lecture says $1 + 2 + 2^2 + \ldots$ "gives -1". It diverges: $\frac{1}{1-x} = 1 + x + x^2 + \ldots$ only for $|x| < 1$; $\frac{1}{1-2} = -1$ is outside that range. ADDED FOR CLARITY T007 00:07:52

Exam trap. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$ has plain denominators; slide 007·5 shows $3!$ and $5!$, which belong to $\sin x$. ADDED FOR CLARITY S007·5

Worked examples

W2.1 · Maclaurin's series of e^x from the definition — ESA ★

GIVEN $f(x) = e^x$.

STEP 1 $f = f' = f'' = \cdots = e^x$, so every value at 0 is 1.

STEP 2 Substitute into E2.2: $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$

ANSWER $$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

CHECK Replacing x by $-x$ gives $e^{-x} = 1 - x + \frac{x^2}{2!} - \ldots$

T007 00:09:10 · T007 00:09:44

SHORT-NOTE RECAP

Taylor about a: $f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \ldots$; Maclaurin about 0. Needs finite derivatives at the point. Standard: e^x, $\log(1+x)$, $(1+x)^n$, $\sin$, $\cos$, $\sinh$, $\cosh$, $\tan^{-1}$, $\sin^{-1}$, $\cos^{-1}$; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.

3 Taylor's expansion of a function of two variables

About a point (a, b): $f(x,y) = f(a,b) + \frac{1}{1!}\{(x-a)f_x + (y-b)f_y\} + \frac{1}{2!}\{(x-a)^2 f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2 f_{yy}\} + \ldots$, with every derivative evaluated at (a, b). The coefficients 1, 2, 1 (and 1, 3, 3, 1 at third order) are binomial, because the mixed partials are equal. ESA questions give the point and the degree.

Key terms and definitions

Taylor polynomial of degree n. The terms of Taylor's series up to total degree n in $(x-a)$ and $(y-b)$; "expand up to second (third) degree terms" means stop there. Degree 1 is the tangent plane.

ADDED FOR CLARITY S007·11 · T007 00:20:13

> *"Taylor's series of $f(x,y)$ about (a,b) is given by $f(x,y) = f(a,b) + \frac{1}{1!}\{(x-a)f_x(a,b) + (y-b)f_y(a,b)\} + \frac{1}{2!}\{(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\} + \ldots$"* S007·9 · T007 00:18:45

Formulas

E3.1 · Taylor's series, two variables (to second order) ESA ★★★

$$f(x,y) = f + \frac{1}{1!}\{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \ldots, \quad h = x-a,\ k = y-b$$

GIVES A polynomial form of f near (a, b).

APPLIES All derivatives evaluated at (a, b), all finite there.

NOTE h, k are shorthand added for clarity; the slide writes $(x-a)$, $(y-b)$.

S007·9 · T007 00:18:07 · T007 00:18:45

E3.2 · Third-order term ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES The terms needed for "up to third degree".

APPLIES Add to E3.1 (or E4.1 with $h = x$, $k = y$).

NOTE Not on the slides; asked in four of six ESA Taylor questions.

ADDED FOR CLARITY ESA Dec-23 1b · ESA Jan-24 s1 1b · ESA Jan-24 s2 1b · ESA Jun-24 1c

E3.3 · Operator form ESA ★★

$$f(a+h,b+k)=\sum_{n=0}^{\infty}\frac{1}{n!}\Big(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\Big)^{n}f(a,b)$$

GIVES Why the coefficients are binomial: 1, 2, 1 and 1, 3, 3, 1.

APPLIES Mixed partials equal ($f_{xy}=f_{yx}$).

NOTE Expand $(h\partial_x+k\partial_y)^n$ like $(h+k)^n$.

ADDED FOR CLARITY T007 00:18:26

Graphs and diagrams

blue: $f(x,y)=xe^{y}+1$

grey: $L(x,y)=1+x+y$

(first degree)

point $(1,0,2)$

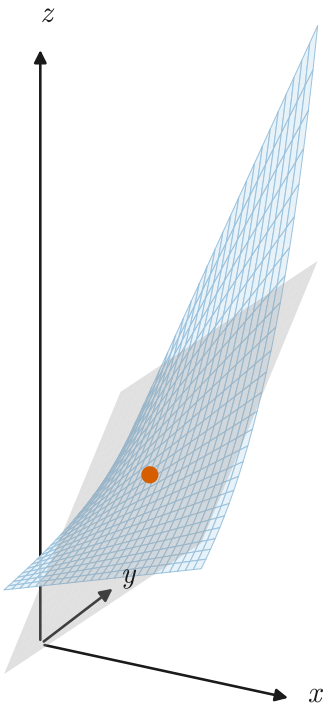


Figure G3.1. $xe^{y}+1$ and its first-degree Taylor polynomial. ESA ★

S007·11 · T007 00:20:13

How to read it. The surface $z=xe^{y}+1$ (blue mesh) and the plane $z=1+x+y$ (grey) touch at $(1,0,2)$. The first-degree Taylor polynomial is the tangent plane: it matches the surface's height and slopes at the point and separates from it away from the point.

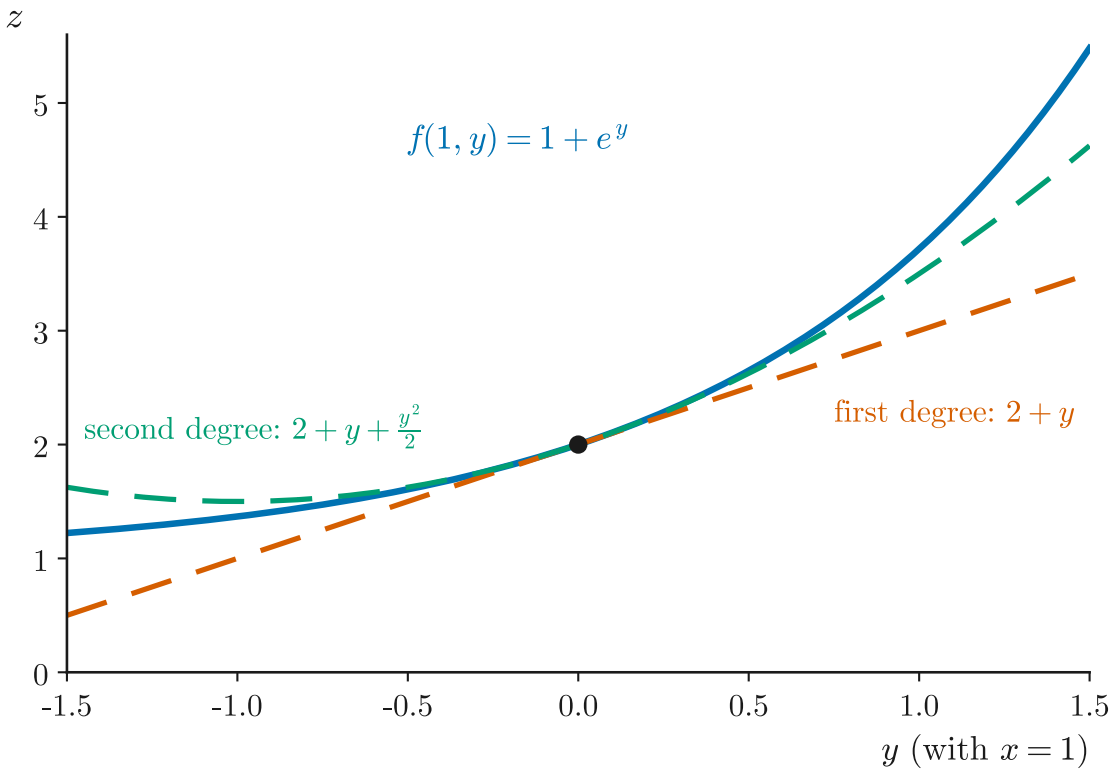
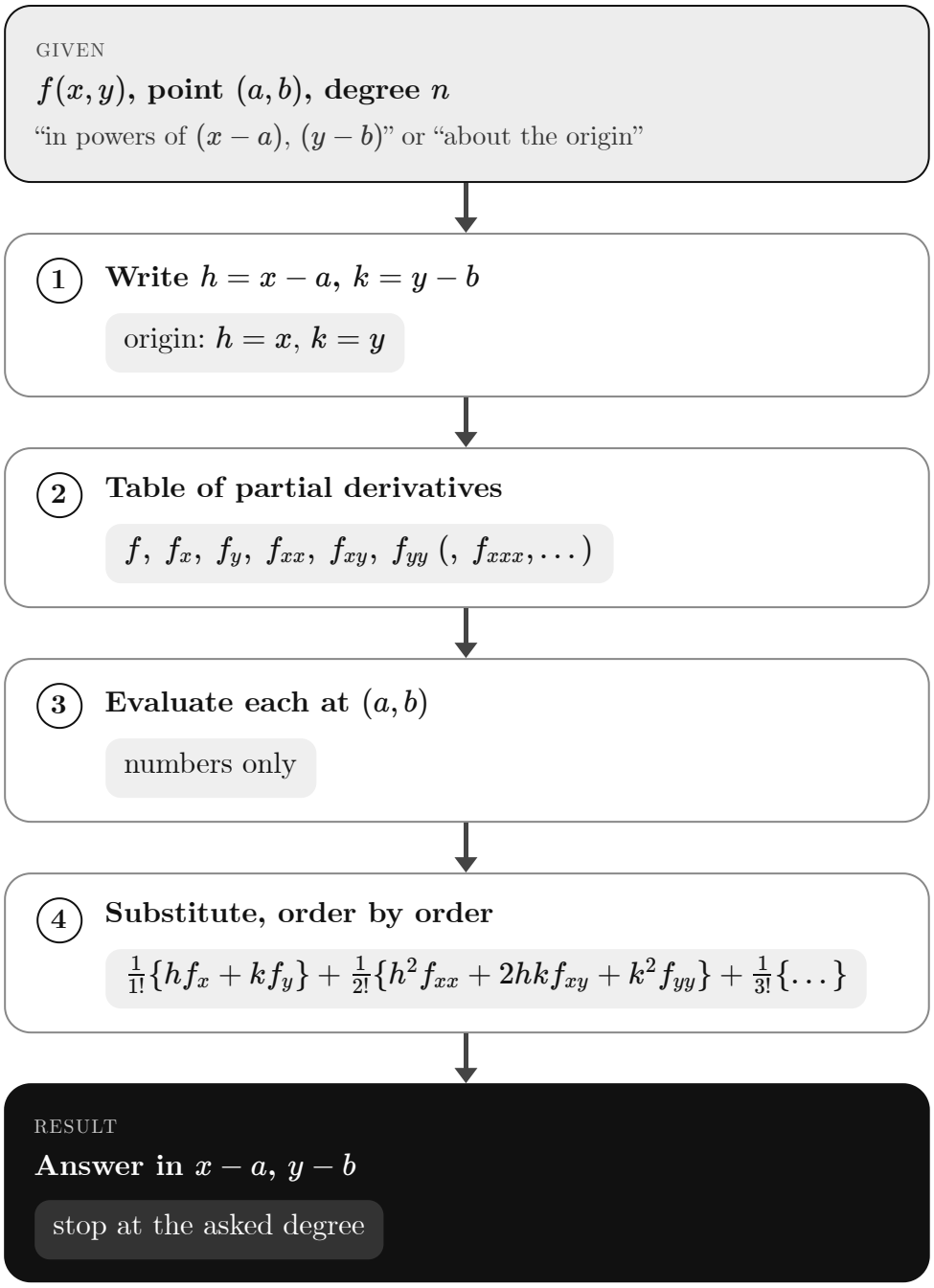


Figure G3.2. Along the line $x = 1$: first- vs second-degree polynomial. ESA ★

ADDED FOR CLARITY S007·11 · T007 00:20:51

How to read it. A slice of Figure G3.1 at $x = 1$: $f(1, y) = 1 + e^y$ (blue), the first-degree polynomial $2 + y$ (vermilion) and the second-degree $1 + x + xy + \frac{y^2}{2} = 2 + y + \frac{y^2}{2}$ (green). The second degree adds curvature and stays close much longer.

Flowcharts



Remark. Check: if f depends only on $x + 2y$, $x - y$, ..., the one-variable series gives the same answer.

Figure F3.1. Expanding f(x, y) in Taylor's series ESA ★★★

S007·9 · ESA Jul-23 1b · ESA Dec-23 1b

Important points and why

1. Lay out a table: f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} (and third order if asked), each in general and then its value at (a, b). Then substitute into E3.1. ESA ★★★ ADDED FOR CLARITY CW U1 P24 · ESA Jul-23 1b

2. There are four second-order partials, but $f_{xy} = f_{yx}$, so they combine into $2(x-a)(y-b)f_{xy}$. ESA ★★ T007 00:18:26
3. "In powers of $(x-1)$ and $(y-\frac{\pi}{4})$" means about the point $(1, \frac{\pi}{4})$. ESA ★★ ADDED FOR CLARITY ESA May-24 1b · CW U1 P27
4. Taylor's series gives approximate values: $\tan^{-1}\frac{0.9}{1.1}$ from the expansion of $\tan^{-1}\frac{y}{x}$ about $(1,1)$ with $h = 0.1$, $k = -0.1$. ESA ★★ CW U1 P26
5. The first-degree Taylor polynomial is the tangent plane; the second degree hugs the surface much better (slide 007·11, Figures G3.1–G3.2). ESA ★ S007·11 · T007 00:20:51

Why are the second-order coefficients 1, 2, 1?
Put $g(t) = f(a+th, b+tk)$ and use the one-variable Maclaurin series of g at $t=0$. By the chain rule $g'(0) = hf_x + kf_y$ and $g''(0) = h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}$ (the two equal mixed partials give the 2). Put $t = 1$.
ADDED FOR CLARITY T007 00:18:26

Exam trap. Evaluate every derivative at (a,b) before substituting; never leave $f_{xx}(x,y)$ in the answer. ADDED FOR CLARITY S007·9

Exam trap. Don't forget $\frac{1}{2!}$ and $\frac{1}{3!}$, and the middle coefficients 2 and 3: the third-order bracket is $h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}$. ADDED FOR CLARITY ESA Dec-23 1b

Exam trap. Slide 007·11's polynomials ($1 + x + y$, $1 + x + xy + \frac{y^2}{2}$) are about $(1,0)$, not the origin; about the origin they are $1 + x$ and $1 + x + xy$. ADDED FOR CLARITY S007·11 · T007 00:20:32

Worked examples

W3.1 · sin x sin y about (π/4, π/4), second order — ESA ★★★

GIVEN Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second-order terms about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$. Let $h = x - \frac{\pi}{4}$, $k = y - \frac{\pi}{4}$.

STEP 1 $f = \sin x \sin y = \frac{1}{2}$; $f_x = \cos x \sin y = \frac{1}{2}$; $f_y = \sin x \cos y = \frac{1}{2}$.

STEP 2 $f_{xx} = -\sin x \sin y = -\frac{1}{2}$; $f_{xy} = \cos x \cos y = \frac{1}{2}$; $f_{yy} = -\sin x \sin y = -\frac{1}{2}$.

STEP 3 E3.1: $\frac{1}{2} + \frac{1}{2}(h+k) + \frac{1}{2}\{-\frac{1}{2}h^2 + 2\cdot\frac{1}{2}hk - \frac{1}{2}k^2\}$.

ANSWER $\boxed{\sin x \sin y \approx \frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}\left(h^2 - 2hk + k^2\right)}$

CHECK CW answer agrees; sympy series agrees.

ESA Jul-23 1b · CW U1 P24

W3.2 · sin x sin y about (π/4, π/4), third degree ESA ★★★

GIVEN Expand $\sin x \sin y$ up to third-degree terms about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

STEP 1 Second order as in W3.1.

STEP 2 $f_{xxx} = -\cos x \sin y = -\frac{1}{2}$, $f_{xxy} = -\sin x \cos y = -\frac{1}{2}$, $f_{xyy} = -\frac{1}{2}$, $f_{yyy} = -\frac{1}{2}$.

STEP 3 E3.2: $\frac{1}{6} \cdot \left(-\frac{1}{2}\right)\{h^3 + 3h^2k + 3hk^2 + k^3\} = -\frac{1}{12}(h+k)^3$.

ANSWER $\boxed{\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3}$

CHECK sympy series to third order agrees term by term.

ESA Dec-23 1b

W3.3 · $e^x \cos y$ about $(1, \frac{\pi}{4})$ ESA ★★★

GIVEN Expand $e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ up to second-degree terms. Let $h = x - 1$, $k = y - \frac{\pi}{4}$.

STEP 1 At $(1, \frac{\pi}{4})$: $f = f_x = f_{xx} = \frac{e}{\sqrt{2}}$ $(e^x \cos y)$; $f_y = f_{xy} = -\frac{e}{\sqrt{2}}$ $(-e^x \sin y)$; $f_{yy} = -\frac{e}{\sqrt{2}}$ $(-e^x \cos y)$.

STEP 2 E3.1: $\frac{e}{\sqrt{2}}\left[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)\right]$.

ANSWER $\boxed{e^x \cos y \approx \frac{e}{\sqrt{2}}\left[1 + (x-1) - \left(y - \frac{\pi}{4}\right) + \frac{1}{2}\left\{(x-1)^2 - 2(x-1)\left(y - \frac{\pi}{4}\right) - \left(y - \frac{\pi}{4}\right)^2\right\}\right]}$

CHECK sympy series agrees.

ESA May-24 1b

W3.4 · √(1+x+y²) in powers of (x−1) and y ESA ★★

GIVEN Expand $\sqrt{1 + x + y^2}$ about $(1, 0)$ up to second degree. Let $h = x - 1$, $k = y$.

STEP 1 $f = \sqrt{2}$; $f_x = \frac{1}{2\sqrt{1+x+y^2}} = \frac{\sqrt{2}}{4}$; $f_y = \frac{y}{\sqrt{\cdots}} = 0$.

STEP 2 $f_{xx} = -\frac{1}{4(1+x+y^2)^{3/2}} = -\frac{\sqrt{2}}{16}$; $f_{xy} = 0$; $f_{yy} = \frac{1+x}{(1+x+y^2)^{3/2}} = \frac{\sqrt{2}}{2}$.

STEP 3 E3.1: $\sqrt{2} + \frac{\sqrt{2}}{4}h + \frac{1}{2}\left(-\frac{\sqrt{2}}{16}h^2 + \frac{\sqrt{2}}{2}k^2\right)$.

ANSWER $\boxed{\sqrt{2}\left[1 + \frac{x-1}{4} - \frac{(x-1)^2}{32} + \frac{y^2}{4}\right]}$

CHECK CW answer agrees; sympy agrees.

CW U1 P27

W3.5 · $xy^2 + y\cos(x-y)$ about $(1,1)$ ESA ★★

GIVEN Expand up to second order about $(1,1)$. Let $h = x-1$, $k = y-1$.

STEP 1 $f = 2$, $f_x = y^2 - y\sin(x-y) = 1$, $f_y = 2xy + \cos(x-y) + y\sin(x-y) = 3$.

STEP 2 $f_{xx} = -y\cos(x-y) = -1$, $f_{xy} = 2y - \sin(x-y) + y\cos(x-y) = 3$, $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y) = 1$.

STEP 3 E3.1: $2 + (h+3k) + \frac{1}{2}(-h^2 + 6hk + k^2)$.

ANSWER $\boxed{2 + [(x-1) + 3(y-1)] + \frac{1}{2}[-(x-1)^2 + 6(x-1)(y-1) + (y-1)^2]}$

CHECK CW answer agrees; sympy agrees.

CW U1 P28

W3.6 · Approximate $\tan^{-1}(0.9/1.1)$ ESA ★★

GIVEN Using $f(x,y) = \tan^{-1}\frac{y}{x}$ about $(1,1)$, compute $\tan^{-1}\frac{0.9}{1.1}$.

STEP 1 At $(1,1)$: $f = \frac{\pi}{4}$, $f_x = \frac{-y}{x^2+y^2} = -\frac{1}{2}$, $f_y = \frac{x}{x^2+y^2} = \frac{1}{2}$, $f_{xx} = \frac{1}{2}$, $f_{xy} = 0$, $f_{yy} = -\frac{1}{2}$.

STEP 2 $\tan^{-1}\frac{y}{x} \approx \frac{\pi}{4} - \frac{1}{2}h + \frac{1}{2}k + \frac{1}{4}h^2 - \frac{1}{4}k^2$.

STEP 3 $h = 0.1$, $k = -0.1$: $0.7854 - 0.05 - 0.05 + 0.0025 - 0.0025$.

ANSWER $\boxed{\tan^{-1}\dfrac{0.9}{1.1} \approx 0.6854}$

CHECK Exact: 0.6857. The CW sheet prints 0.6904, an arithmetic slip.

CW U1 P26

W3.7 · The slide's $xe^y + 1$ polynomials ESA ★

GIVEN Find the first- and second-degree Taylor polynomials of $f = xe^y + 1$ about $(1,0)$.

STEP 1 $f = 2$, $f_x = e^y = 1$, $f_y = xe^y = 1$; $f_{xx} = 0$, $f_{xy} = e^y = 1$, $f_{yy} = xe^y = 1$.

STEP 2 First degree: $2 + (x-1) + y = 1 + x + y$.

STEP 3 Second degree: add $\frac{1}{2}\{2(x-1)y + y^2\}$, giving $1 + x + xy + \frac{y^2}{2}$.

ANSWER $\boxed{1 + x + y \text{ and } 1 + x + xy + \dfrac{y^2}{2}}$

CHECK These are slide 007·11's polynomials, so the slide expands about $(1,0)$.

ADDED FOR CLARITY S007·11 · T007 00:20:13

Past-paper questions

ESA Jul 2023 7 marks UE22MA141A Q1.b · solved in W3.1

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W3.2

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to third degree term about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W3.3

Expand $f(x,y) = e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ using Taylor's series up to second degree terms.

SHORT-NOTE RECAP

About (a,b) with $h = x-a$, $k = y-b$: $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$, all at (a,b). ESA: $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$ gives $\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3$.

4 Maclaurin's expansion of a function of two variables

Put $(a,b)=(0,0)$: $f(x,y)=f(0,0)+\frac{1}{1!}\{xf_x+yf_y\}+\frac{1}{2!}\{x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\}+\ldots$ at the origin. f_x goes with x, f_y with y, f_{xx} with x^2, f_{xy} with $2xy$, f_{yy} with y^2. When f is a function of $x+2y$ or $x-y$, the one-variable series gives the answer as a check.

Key terms and definitions

Maclaurin's series (two variables). Taylor's series of $f(x,y)$ about the origin $(0,0)$. S007·10 · T007 00:19:01

> *"Maclaurin's series for a function of two variables is given by* $f(x,y)=f(0,0)+\frac{1}{1!}\{xf_x(0,0)+yf_y(0,0)\}+\frac{1}{2!}\{x^2f_{xx}(0,0)+2xyf_{xy}(0,0)+y^2f_{yy}(0,0)\}+\ldots$*"* S007·10

Formulas

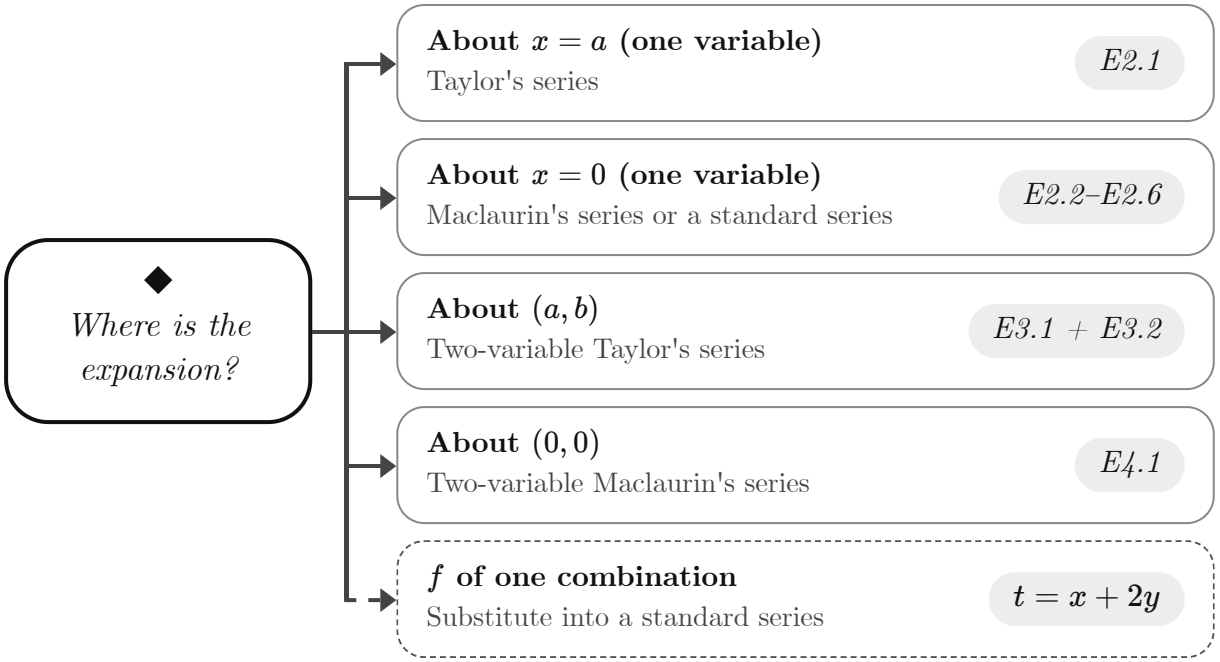
E4.1 · Maclaurin's series, two variables ESA ★★★

$$f(x,y)=f(0,0)+\frac{1}{1!}\{xf_x+yf_y\}+\frac{1}{2!}\{x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\}+\ldots$$

gives	A polynomial form of f near the origin.
applies	Derivatives at $(0,0)$, all finite there.
note	Same as E3.1 with $a=b=0$; add E3.2 with $h=x$, $k=y$ for third degree.

S007·10 · T007 00:19:38

Flowcharts



Remark. The dashed route is a shortcut added for clarity; it is also the quickest way to check an answer.

Figure F4.1. Which series? ESA ★★

T007 00:11:17 · S007·9 · S007·10

Taylor's and Maclaurin's series side by side

	One variable	Two variables
Taylor (about a point)	$f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$	$f(a,b) + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots$
Maclaurin (about the origin)	$f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots$	$f(0,0) + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \dots$
First degree	Tangent line	Tangent plane
Coefficients at order n	$\frac{1}{n!}$	$\frac{1}{n!}\times$ binomial (1,2,1; 1,3,3,1)
Condition	f and derivatives finite at the point	f and all partials finite at the point

ADDED FOR CLARITY S007·4 · S007·9 · S007·10

Important points and why

1. Memory aid from the lecture: f_x goes with x, f_y with y, f_{xx} with x^2, f_{xy} with $2xy$, f_{yy} with y^2. ESA ★★ T007 00:19:38
2. Shortcut and check: if f depends on one combination, use the one-variable series: $\sin(x+2y) = (x+2y) - \frac{(x+2y)^3}{6} + \ldots$, $\log(1+x-y) = (x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3} - \ldots$ ESA ★★★
   ADDED FOR CLARITY CW U1 P25 · ESA Jun-24 1c · ESA Jan-24 s1 1b
3. None of f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} may be infinite at $(0,0)$. ESA ★ T007 00:19:26

Worked examples

W4.1 · log(1 + x − y) about the origin — ESA ★★★

GIVEN Expand $\log(1+x-y)$ about the origin up to third-degree terms.

STEP 1 At $(0,0)$: $f=0$; $f_x = \frac{1}{1+x-y} = 1$, $f_y = -1$.

STEP 2 $f_{xx} = -1$, $f_{xy} = 1$, $f_{yy} = -1$.

STEP 3 $f_{xxx} = 2$, $f_{xxy} = -2$, $f_{xyy} = 2$, $f_{yyy} = -2$.

STEP 4 E4.1 + E3.2: $(x-y) + \frac{1}{2}(-x^2+2xy-y^2) + \frac{1}{6}(2x^3 - 6x^2y + 6xy^2 - 2y^3)$.

ANSWER $\boxed{(x-y) - \dfrac{(x-y)^2}{2} + \dfrac{(x-y)^3}{3}}$

CHECK The same as $\log(1+t)$ with $t = x-y$; sympy agrees.

ESA Jan-24 s1 1b

W4.2 · $e^{ax}\sin by$ about the origin — ESA ★★★

GIVEN Expand $e^{ax}\sin by$ about the origin up to third-degree terms.

STEP 1 At $(0,0)$: $f=0$, $f_x = ae^{ax}\sin by = 0$, $f_y = be^{ax}\cos by = b$.

STEP 2 $f_{xx} = 0$, $f_{xy} = ab$, $f_{yy} = 0$.

STEP 3 $f_{xxx} = 0$, $f_{xxy} = a^2 b$, $f_{xyy} = 0$, $f_{yyy} = -b^3$.

STEP 4 E4.1 + E3.2: $by + \frac{1}{2}(2abxy) + \frac{1}{6}(3a^2bx^2y - b^3y^3)$.

ANSWER $\boxed{by + abxy + \dfrac{a^2 b\, x^2 y}{2} - \dfrac{b^3 y^3}{6}}$

CHECK Product of $1 + ax + \frac{a^2x^2}{2}$ and $by - \frac{b^3y^3}{6}$, keeping degree ≤ 3; sympy agrees.

ESA Jan-24 s2 1b

W4.3 · sin(x + 2y) about the origin ESA ★★★

GIVEN Expand $\sin(x+2y)$ in Taylor's series up to third-degree terms about $(0,0)$.

STEP 1 $f=0,\ f_x=\cos(x+2y)=1,\ f_y=2.$

STEP 2 $f_{xx}=-\sin(\cdot)=0,\ f_{xy}=0,\ f_{yy}=0.$

STEP 3 $f_{xxx}=-1,\ f_{xxy}=-2,\ f_{xyy}=-4,\ f_{yyy}=-8.$

STEP 4 $x+2y+\frac{1}{6}\{-x^3-6x^2y-12xy^2-8y^3\}.$

ANSWER $\boxed{(x+2y)-\dfrac{(x+2y)^3}{6}}$

CHECK CW answer agrees; the sine series with $t=x+2y$ gives the same.

ESA Jun-24 1c · CW U1 P25

Past-paper questions

ESA Jan 2024 · 7 marks · UE23MA141A set 1, Q1.b · solved in W4.1

Expand $\log(1+x-y)$ about origin up to third degree terms.

ESA Jan 2024 · 7 marks · UE23MA141A set 2, Q1.b · solved in W4.2

Expand $e^{ax}\sin by$ about origin up to third degree terms.

ESA June 2024 · 6 marks · UE23MA141A Q1.c · solved in W4.3

Expand $f(x,y)=\sin(x+2y)$ in Taylor's series up to third degree terms about the point $(0,0)$.

SHORT-NOTE RECAP

About $(0,0)$: $f+\{xf_x+yf_y\}+\frac{1}{2!}\{x^2f_{xx}+2xyf_{xy}+y^2f_{yy}\}+\ldots$ ESA: $\log(1+x-y)=(x-y)-\frac{(x-y)^2}{2}+\frac{(x-y)^3}{3}$; $e^{ax}\sin by=by+abxy+\frac{a^2b}{2}x^2y-\frac{b^3}{6}y^3$; $\sin(x+2y)=(x+2y)-\frac{(x+2y)^3}{6}$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
a, (a, b)	Expansion point	Point about which the series is written	E2.1, E3.1
$h,\ k$	Increments	$h = x - a$, $k = y - b$ (shorthand added for clarity)	E3.1–E3.3
$f^{(n)}(a)$	n-th derivative	Printed $f^n(a)$ on slide 007·4; not a power	E2.1
$n!$	Factorial	$1 \cdot 2 \cdots n$; divides the n-th order term	E2.1–E4.1
$f_x,\ f_y$	First partials	At the expansion point	E3.1, E4.1
$f_{xx},\ f_{xy},\ f_{yy}$	Second partials	$f_{xy} = f_{yx}$, hence the 2	E3.1, E4.1
$f_{xxx}, \ldots, f_{yyy}$	Third partials	Coefficients 1, 3, 3, 1	E3.2
n (in $(1+x)^n$)	Exponent	Any real number; not the order of a derivative	E2.4
$L(x, y)$	Taylor polynomial	Slide 007·11's name for the truncated series	W3.7

B Sources, coverage and gaps

Source	Path	Use
Deck 007 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.*	13 slides; content on 3–12
Transcript 007 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.*	22.2 min lecture
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 24–28 (with answers)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Video 007 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_*.mp4	Not opened: every slide is readable, and the one garbled formula is read out in full in the transcript

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A Taylor or Maclaurin expansion in two variables is asked in six of them, always as a 6–7 mark Q1 part. About a point (a, b): Dec 2023 Q1.b ($\sin x \sin y$, third degree), Jul 2023 Q1.b (the same function, second order) and May 2024 Q1.b ($e^x \cos y$ about $(1, \pi/4)$). About the origin: Jan 2024 set 1 Q1.b ($\log(1 + x - y)$), set 2 Q1.b ($e^{ax} \sin by$) and June 2024 Q1.c ($\sin(x + 2y)$). Four of the six ask for third-degree terms, which the slides do not write out. Jan 2024 sets 3 and 4 have no Taylor question. One-variable series and the applications were never asked.

Known gaps, conflicts and corrections

1. Slide 007·9's second-order term is garbled in the rendering. It was rebuilt from the transcript (T007 00:18:45) and the Maclaurin form on slide 007·10: $\frac{1}{2!}\{(x-a)^2 f_{xx} + 2(x-a)(y-b) f_{xy} + (y-b)^2 f_{yy}\}$ at (a, b).
2. The slides stop at second order. Four of the six ESA questions ask for third-degree terms, so the third-order term $\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$ and the operator form are added for clarity.
3. Slide 007·3 asks for $\lim_{x \to 0} \frac{\sin x - x}{x^4}$. That limit does not exist (it behaves like $-\frac{1}{6x}$); with x^3 in the denominator the limit is $-\frac{1}{6}$, worked as W1.2.
4. Slide 007·3's integrals ($\int e^{x^2} dx$, $\int \cos(x^2) dx$) are partly lost in the rendering; the lecture uses $\int_0^1 \sin(x^2)\, dx$ instead (T007 00:11:51), worked as W1.1.
5. Slide 007·7 says "$E = mc^2$ is an approximation for low velocities". More precisely, the Maclaurin expansion of the relativistic energy γmc^2 gives $mc^2 + \frac{1}{2}mv^2$ for $v \ll c$ (added for clarity).

6. Slide 007·8's graphs of $\sin x$ and its Maclaurin polynomials have no axis labels; they are redrawn as Figure G1.1.
7. Slide 007·11 does not state the expansion point. Its polynomials $1 + x + y$ and $1 + x + xy + \frac{y^2}{2}$ are the Taylor polynomials of $xe^y + 1$ about $(1, 0)$; the lecture's guess that "maybe they have used the Maclaurin series" (T007 00:20:32) is wrong: about the origin they would be $1 + x$ and $1 + x + xy$ (sympy).
8. Slide 007·12 is titled "Applications of Euler's Theorem of a multivariate function"; its content is the applications of Taylor's series (a copy error).
9. The lecture says $1 + 2 + 2^2 + 2^3 + \ldots$ "gives -1" (T007 00:07:52). The series diverges: $\frac{1}{1-x} = 1 + x + x^2 + \ldots$ holds only for $|x| < 1$. Corrected as an exam trap.
10. The lecture's condition "$f(a)$ cannot tend to infinity" (T007 00:10:01) is incomplete: f needs derivatives of every order at the point, and the series must converge to f. The precise condition is added for clarity.
11. The lecture's picture of a calculator summing Maclaurin series is a simplification; real calculators use several polynomial and shift-and-add methods. It is kept as the lecture's motivation.
12. Class-work problem 26 prints $\tan^{-1}(0.9/1.1) \approx 0.6904$. The second-order Taylor value is 0.6854 (exact 0.6857).
13. Class 008 ("Problems on Taylor's and Maclaurin's series") is a separate session and is not included.
14. Transcript recognition errors were corrected silently: "McLaren" / "Mechlerin" / "Makelden" / "metogen" → Maclaurin, "McLevins" → Maclaurin, "t-shirts" → Taylor series, "L'opitholes" → L'Hôpital's, "Legrange" → Lagrange, "four-year series" → Fourier series, "sinex" → sin x, "2a into" → 2(x−a)(y−b).

Coverage

Source	Citations	Slides cited (slide × citations)
S007	35	3×2, 4×5, 5×6, 6×1, 7×1, 8×2, 9×6, 10×5, 11×6, 12×1
CW	9	
ESA	17	
T007	49	

Slides 007·1, 007·2 and 007·13 are title, session and closing slides with no content.